D 145 | Follow the cash, then the contract

PRIVATE FICTIONAL offline lab. All money millions. No distribution instruction, actual fund returns, legal/tax advice, account, commitment or transfer.

Capital first	Commitments 60; LP 50 paid at time 0, 40 invested and 10 fees/expenses inside commitment, 10 undrawn. No GP capital, debt, recycling, tax, other cash flows or NAV. Fees are already inside 50, not deducted again from 100 available cash at exactly year 5.
Stipulated full catch-up	Return LP capital first, then 8% annual compound preference on 50 for 5 whole years. Then 100% GP catch-up until GP has 20% of profit distributed through preference/catch-up; split remainder 80/20. Not the ILPA model bracketed 80% GP tapered catch-up, not a hard hurdle or universal term.
Cash is a ceiling	At 100 available: preference target 23.46640384, catch-up target 5.86660096, residual 20.6669952, LP 90/GP 10. At 75, only 1.53359616 available for catch-up and zero residual; LP 73.46640384/GP 1.53359616. At 30, LP 30, capital shortfall 20, no pref or carry.
Ratios and time	Base LP distributions 90/paid 50=net DPI 1.8, NAV 0 so same TVPI. Portfolio 100/invested 40=2.5 is a different basis, not reported ILPA fund-level gross MOIC. Exactly two investor flows/time 0 and year 5 permit synthetic annual endpoint IRR 1.8^(1/5)-1; no inference from undated multiples.
Separate early carry toy	A 20 cost realizes 60 year 2; B 20 cost realizes 0 year 5; 40 total paid at time 0, no fees/preference/NAV. Early deal carry 8; final whole-fund entitlement 4; stipulated gross true-up 4 due. Zero assumed actual recovery unless evidence says otherwise. Shortfall is not forgiven or guaranteed collectible.
Try first	Change cash 100 to 75, capital fees, preference, years, carry and recovery. Explain each exhausted tier. Scenario grid varies available cash only. Static prose gives baseline reference, not new actual performance. Five-row actual evidence/authority/owner/stopplan remains blank.

Inputs | Two separate stipulated families

Money millions. Rates fractions. Annual compound preference on full paid-in at time 0; one endpoint only. No partial periods or intermediate flows.

Fund commitments	60.000000
LP paid-in capital at time 0, all inside commitment	50.000000
Invested portfolio cost from paid-in	40.000000
Fees/expenses from paid-in	10.000000
AFTER-ALL-COST cash available at endpoint	100.000000
Annual compound preferred return fraction	0.080000
Whole years to the single endpoint	5.000000
Carry fraction, full 100%GP catch-up stipulated	0.200000

Separate timing toy A investment cost	20.000000
Separate timing toy A cash realization year 2	60.000000
Separate timing toy B investment cost	20.000000
Separate timing toy B cash realization year 5	0.000000
Separate timing toy carry fraction	0.200000
ASSUMED actual clawback recovery fraction	0.000000

Capital and waterfall numeric gate	1.000000
Timing and recovery numeric gate	1.000000

Numeric gates do not establish a valid contract. Paid-in must equal invested+fees and not exceed commitments in this toy. 100% carry excluded: full catch-up target has denominator 1-carry. Timing family independent. Recovery fraction 0 is an assumption, not observed default.

Waterfall | Cash-capped 100%GP catch-up

Millions except labeled ratios/rates. Cash-capped authored arithmetic, not an actual entitlement or certified fund report.

Undrawn commitment	10.000000
Capital paid to LP first	50.000000
Cash after capital	50.000000
Compound preference target, not guaranteed	23.466404
Preference actually paid to LP	23.466404
Cash after preference	26.533596
Full catch-up target on preference target	5.866601
Catch-up actually paid to GP	5.866601
Residual cash after capped catch-up	20.666995
LP residual	16.533596
GP residual	4.133399
Total LP distribution	90.000000
Total GP carry distribution	10.000000
Cash reconciliation, must be 0	0.000000
Unreturned capital	0.000000
Unpaid preference target	0.000000

Base LP 90/GP 10, sum 100; preference 23.46640384, catch-up 5.86660096, residual 20.6669952. At 75: LP 73.46640384, GP 1.53359616, residual 0, unpaid catch-up not promised future cash. At 30: capital shortfall 20, pref unpaid 23.46640384, carry 0. Full catch-up, not a hard-hurdle carry only on excess profit. No negative cash tier.

Ratios | State denominator and dates

Millions except labeled ratios/rates. Cash-capped authored arithmetic, not an actual entitlement or certified fund report.

Net DPI, LP distribution / LP paid-in	1.800000
Net TVPI, NAV stipulated zero	1.800000
Gross portfolio proceeds / portfolio cost	2.500000
Synthetic LP two-flow endpoint annual IRR	0.124746
Synthetic portfolio two-flow annual rate	0.201124
Unpaid catch-up target, not a payable promise	0.000000

Baseline DPI/TVPI 1.8, portfolio multiple 2.5, LP annual endpoint rate about 12.4746%, portfolio about 20.1124%. No residual value, interim flows, GP investment, fees after time 0, facility or tax. Portfolio multiple is not ILPA fund-level gross MOIC. Changing dates/flows requires a dated cash-flow model, not this power formula. Zero distributions leave this IRR undefined, not a valid -100% root.

Timing | Early carry is not final entitlement

Millions except labeled ratios/rates. Cash-capped authored arithmetic, not an actual entitlement or certified fund report.

Total separate capital contributions at time 0	40.000000
Total separate realized cash	60.000000
Early carry on A profit only	8.000000
Whole-fund final profit, nonnegative base	20.000000
Stipulated ultimate whole-fund entitlement	4.000000
Gross excess carry due back under stipulation	4.000000
Assumed actual gross recovery	0.000000
Unrecovered gross amount due	4.000000
Carry retained after assumed recovery	8.000000
Capital providers cash after assumed recovery	52.000000
Whole-fund ultimate provider entitlement	56.000000
Provider gap to ultimate entitlement	4.000000
Additional carry if ultimate exceeds early	0.000000

Base early 8, final 4, gross due 4, assumed recovered 0, unrecovered 4, carry retained 8, providers 52 versus ultimate 56, gap 4. Recovery 100% gives retained 4/providers 56. If laterB profitable enough, additional carry rather than clawback may be due; no later carry paid in this toy. No escrow, tax cap, solvency, enforcement, guarantor, time-value or legally collectible amount estimated.

Cash grid | Exhaust each tier in order

Only available cash varies. Shared assumptions from Inputs. Formula outputs not a forecast, probability, sensitivity distribution or signed entitlement.

Cash INPUT	Capital LP	Pref LP	Catch-up GP	Residual cash	Total LP	Total GP
0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
30.000000	30.000000	0.000000	0.000000	0.000000	30.000000	0.000000
50.000000	50.000000	0.000000	0.000000	0.000000	50.000000	0.000000
60.000000	50.000000	10.000000	0.000000	0.000000	60.000000	0.000000
75.000000	50.000000	23.466404	1.533596	0.000000	73.466404	1.533596
100.000000	50.000000	23.466404	5.866601	20.666995	90.000000	10.000000
150.000000	50.000000	23.466404	5.866601	70.666995	130.000000	20.000000

Row cash values are independent scenario inputs, not linked to Inputs cash. Zero cash returns zero, not negative carry. Each cash-capped stage stops when cash is exhausted. Fee 10 was already funded within 50, not subtracted again. Inputs gate ties all rows to complete main numeric family.

Contract | Read the operative clause

Authored comparisons, no actual LPA interpretation, investor preference, current-law conclusion or distribution approval.

What counts as capital	Specify committed versus contributed versus invested, fee/expense base, inside/outside commitment and GP own capital. Toy fees inside 50; ILPA performance definitions allow outside-commitment calls. Do not infer undrawn amount from invested capital alone.
Pref, hard hurdle, catch-up	Specify simple/compound rate, days/calendar, base, timing and any accrual on unpaid preference. Toy one dated endpoint/full 100%GP catch-up. ILPA July 2020 model 14.3.3 instead brackets 80%GP/20%LP during catch-up. Principles 3.0 prefers a hard hurdle. These differ, not one interchangeable convention.
Accrual versus cash	Accounting allocation/accrued carry does not establish a right to cash distribution. European terminology does not eliminate later calls or clawback risk. No universal 8%pref/20%carry/100%catch-up or all-loss protection; capital/preferences can remain unpaid.
Due versus recovered	Clawback trigger, base, testing dates, tax cap, tax benefit, escrow, individual undertakings, enforcement and survival matter. ILPA model 14.7 has detailed lesser/greater tests and tax limit, not this gross 4 toy. Optional 30%escrow is model language, not cash evidence or a universal security guarantee.
Separate measurement scopes	Net investor flows include fee/expense/carry effects. Portfolio cost/proceeds is not fund-level gross performance. NAV, facility impacts and inside/outside commitments change definitions. No universal gross-to-net haircut, guaranteed tax rate, IRR without dates or actual payout instruction.

Actual plan | Five rows remain blank

No actual LPA, distribution notice, capital ledger, bank proof, counsel review, accepted owner or payout authority.

Question / entity	Source / cutoff	Operative clause	Observed / toy	Owner / acceptance	Permission / budget	Stop / delayed transfer

Do not fill with invented evidence. No distribution, clawback demand, solicitation, fund formation, account, commitment or transfer authorized by this paper. Fullsource/AV/native cross-viewer/learner/account/legal/tax/overall 10 remain open.

Evidence | Sources and differing terms

Six URLs, three organizations. Legal commentary, vendor exposition, association model/standards; not six independent studies or a legal opinion.

Cooley primer	https://thefundlawyer.cooley.com/primer-carried-interest-in-venture-capital-funds/ | June 10, 2026. Selected terminology, carry, pref, base, timing, clawback and conclusion read. Tax/1061/waiver/securities sections not reviewed. Surveys and market frequencies not independently verified; no US tax advice or universal terms.
Carta waterfall	https://carta.com/learn/private-funds/management/distribution-waterfall/ | May 28, 2025. Complete returned written article read, not linked downloadable guide or image pixels. Company equity preference and fund waterfalls distinct; promised preference is priority, not guaranteed cash.
ILPA definitions	https://ilpa.org/wp-content/uploads/2025/04/ILPA-Performance-Template-Definitions-Granular-Methodology-v1.1.pdf | Selected calls/commitments, netIRR/netTVPI, grossIRR/gross fundMOIC read, not full PDF. Investor versus portfolio metrics differ. Toy endpoint rate not a full dated investor performance implementation.
ILPA model/overview	https://ilpa.org/wp-content/uploads/2020/07/ILPA-Model-Limited-Partnership-Agreement-WOF.pdf and https://ilpa.org/wp-content/uploads/2020/07/ILPA-Model-LPA-Overview-WOF-Version-1.pdf | July 2020: selected definition, 14.3 and 14.7 downloaded model; complete returned overview. Model is Delaware buyout drafting reference, not Indian law/fullagreementreview.
ILPA principles	https://ilpa.org/wp-content/uploads/2019/06/ILPA-Principles-3.0_2019.pdf | Selected waterfall, carry, recycling and clawback written guidance, not full PDF. Prefers hard hurdle, not toy full catch-up; gross-of-tax recommendation differs from model tax-capped test. Do not silently merge these terms.
What remains	Model catch-up 80/20, principles hardhurdle, and toy 100%GPcatch-up intentionally distinguished. No current law/actual contract, source image/AV, native Excel/Sheets parity, independent learner, account/field/legal/tax/overall 10 clearance. Recovery is separate from amount due.

Failure log | Authored tests only

Populated after execution. Not investor errors, independent review, certified performance or legal approval.

1	PASS: 79 baseline formulas independently recomputed.
2	PASS: 24 changed sets plus restore.
3	PASS: 24 invalid inputs/1896 independent isolation checks.
4	PASS: 6 broken references/6 explicit checks.
5	PASS: main/timing/grid invalid family isolation checked.
6	PASS: 5 boundary sets/395 checks; zero-cash IRR undefined.
7	PASS: amount due distinct from assumed recovery/no legal claim.
8	PASS: actual evidence/permissions/owner/stop plan blank.